Assessment 4: Final Mobile App

Due Date: Thursday Week 14 (6th November, 11:55PM) - Weight: 30%

Purpose:

The purpose of this assessment is to produce a complete iOS application based on the idea described in your App Design Specification (Assessment 1). This assessment tests your ability to apply knowledge from all topics covered throughout the semester

Completion of this assessment should demonstrate the following learning outcomes:

- Analyse mobile interface guidelines and technical constraints to design effective navigation and user interfaces for mobile apps
- Apply common object-oriented design patterns such as Model-View-Controller and Delegation
- Follow iOS best practices to design, construct and test non-trivial iOS apps with a web service component

Task:

For this assessment, you will develop and submit a complete iOS app based on the design submitted for Assessment 1. You should have already demonstrated considerable progress towards the final version of your application through the demonstrations of your app development (Assessment 3).

Your app should be built to a high standard using the knowledge you have gained throughout FIT3178 as well as via further research into iOS development. Your app should demonstrate both a robust implementation of functionality as outlined in Assessment 1 as well as a high-quality user interface design.

You will demonstrate your completed app to your lab demonstrator during a 30-minute interview during Week 15 (the week after the submission deadline). This provides you with the opportunity to showcase your app running in your development environment or personal device(s) and discuss the final source code.

It is expected that you will be able to provide a live demonstration of your application on the day of your presentation. If this is going to be an issue or you have additional requirements for your demonstration, please consult with your lab demonstrator before your demonstration.

Marking Criteria:

This assessment is worth 30% of your total marks for this unit.

Your app will be graded in terms of the overall quality of implemented functionality (the code) and the user experience when using your application (the demonstration). An outline of criteria for this assessment are outlined below:

UI Design – 20 Marks

- Visual UI Design
    - The app interface is aesthetically pleasing and professional
    - UI design is appropriate for the target audience
    - The interface uses the appropriate UI controls and is consistent across the app
- Usability and Navigation
    - All screens of content are accessible from the user interface
    - Errors are handled with informative error messages
    - It is clear how to interact with and use all parts of the app
    - The app follows the advice outlined in the Apple Human Interface Guidelines

Application Functionality – 40 Marks

- Overall Quality of Functionality
    - Key features outlined in the project specification are implemented
    - Features work correctly in the demonstration without issues (no bugs)
    - Exceptions/errors are handled appropriately and don't crash the application at any stage
- Overall Complexity of Functionality
    - A sufficient level of complexity in implemented features is demonstrated
    - All technology requirements from Assessment 1 spec are included
    - Use of API features beyond those shown in lab exercises and workshops

Programming Implementation – 40 Marks

- Overall Code Quality
    - The app shows appropriate use of MVC and OO design
    - The app shows appropriate use of frameworks and patterns (e.g., Delegation, View Controllers, Closures)
    - The code follows appropriate naming conventions for Swift code for class properties. protocols, methods, variables, etc.
- Overall Code Documentation
    - There are well-written, clear comments throughout the code to explain the structure and design
    - Code comments demonstrate a clear understanding of Swift code and utilised iOS frameworks
    - All use of online sources is appropriately referenced
    - <u>The app includes a view with acknowledgements of any included libraries</u>

Submission Requirements:

Your assessment will need to be submitted online via Moodle. Your project should be submitted as a zip archive containing the app code, documentation (and any backend implementation code / datasets, if applicable). Ensure that the archive is named using the following convention STUDENTNAME-A4-FinalApplication.zip.

As this assessment is due during the examination period, you may potentially have multiple examinations scheduled on or near the assessment due date. If so, please contact your lab demonstrator and provide your examination timetable and you may be granted an extension.

Absence and Special Consideration

Failure to submit your assessment on time will result in a 10%-mark penalty for each day late (including weekends) up to a maximum of 7 days late. Submissions later than 7 days will receive a mark of 0.

If you are unable to work on your assessment for a significant period due to circumstances beyond your control, you may be able to apply for special consideration. Special consideration applications must be sent to Monash Connect via the following link https://www.monash.edu/connect/forms/modules/course/special-consideration. Supporting documentation will need to be submitted with the application.

Generative AI tools are restricted for certain functions in this assessment task

In this assessment, you can use generative artificial intelligence (AI) in order to conduct background research only, however, we strongly discourage it as it doesn't give its sources, and it may not give correct answers or best-practice. Any use of generative AI must be appropriately acknowledged (see see Learn HQ) including specifying the prompts used.

Acknowledgment of third-party materials

It is critical that you provide attribution for any external resource used to create your app, even if modified for your own purposes.

- Any open-source library used must be given attribution based upon its license requirements. You can use an About screen in your app to present the libraries used and their license notices.
- Any other external resources that you have used (such as YouTube video tutorials, StackOverflow answers, etc.) must be acknowledged within your app's source code.

Failure to appropriately attribute or acknowledge materials used to develop your application can be considered as plagiarism and may result in a mark of 0 being awarded for the assessment.